Lecture 6.2 – Final Controller Validation and Project Preparation

Week 6: Turning Six Weeks of Work into One Complete Control-System Story

Learning Objectives

By the end of this lecture, you should be able to:

- identify the required evidence for a final controller;
- plan a repeatable final test;
- evaluate tracking, overshoot, saturation, and repeatability;
- troubleshoot common motor-control problems systematically;
- organize the final presentation around measured results.

1. The Complete Course Story

The course has followed one motor from open-loop operation to feedback control.

1. Week 1: open-loop PWM and encoder measurement.
2. Week 2: first-order motor model.
3. Week 3: proportional control.
4. Week 4: PI control.
5. Week 5: digital implementation and filtering.
6. Week 6: final validation and project preparation.

The final project should tell this engineering story using your own motor data.

2. What Makes a Good Final Controller?

A good final controller should:

- reach the desired speed;
- have small steady-state error;
- avoid excessive overshoot;

- avoid long PWM saturation;
- behave smoothly;
- give similar results across repeated trials.

It does not need to be perfect.
It does need to be explainable and supported by data.

3. Final Validation Tests

Recommended final tests:

1. low reference speed;
2. medium reference speed;
3. higher safe reference speed;
4. repeated test at the main reference speed;
5. optional safe disturbance test.

Use instructor-approved speeds that your motor can reach reliably.

4. Example Reference Speeds

If your motor can safely reach approximately 160 RPM, one possible set is:

$$80 \text{ RPM}, \qquad 120 \text{ RPM}, \qquad 150 \text{ RPM}.$$

These are examples only.
The goal is to show that the controller works at more than one operating point.

5. What to Measure

For each final test, record:

Metric	Meaning
Reference RPM	target
Final RPM	achieved speed
Steady-state error	tracking accuracy
Overshoot	excessive response
Maximum PWM	actuator effort
Settling behavior	speed of response
Integral state	windup behavior

6. Repeatability

A controller that works once is less convincing than a controller that works repeatedly.
Run the main-reference test several times.
Compare:

- final speed;
- steady-state error;
- response shape;
- maximum PWM.

Repeatability is evidence that the controller is reliable rather than lucky.

7. Optional Disturbance Test

A disturbance test shows the benefit of feedback.
With instructor approval, apply a small safe load change.
Observe:

- RPM drop;
- error increase;
- PWM response;
- recovery toward the reference.

Do not touch the spinning wheel directly.

8. Troubleshooting Strategy

When something does not work, change one thing at a time.

Symptom	Check first
RPM always zero	encoder wiring / counts
Motor does not turn	power / L298N / PWM
Wrong direction	IN1/IN2 or motor leads
RPM very noisy	sample time / filter / encoder counts
Large final error	K_p, K_i, saturation
Large overshoot	K_i, windup, filtering delay
PWM always 255	reference too high or gains too aggressive

9. Do Not Retune Everything at Once

If the final controller behaves poorly:

1. verify wiring and encoder measurement;
2. verify reference speed is achievable;
3. verify sample time;
4. check PWM saturation;
5. reduce K_i if overshoot or windup is severe;
6. adjust K_p only if necessary;
7. change filter strength only after controller behavior is understood.

10. Required Final Figures

A strong final presentation should include a small number of clear plots.
Recommended:

1. Week 2 measured response and first-order model;
2. P vs. PI control comparison;
3. final controller tracking plot;

4. optional repeated-trial or disturbance plot.

Do not show every plot generated during the course.
Choose plots that support the design story.

11. Suggested Presentation Story

A clear presentation can follow this order:

1. What was the control goal?
2. What hardware did we use?
3. What did open-loop behavior look like?
4. How did we build the motor model?
5. What happened with P control?
6. Why did we add integral action?
7. What final controller did we choose?
8. How well did it work?
9. What would we improve next?

12. Final Controller Table

Record your final settings.

Setting	Final value
Reference RPM	
K_p	
K_i	
Sample time T_s	
Filter α	
Integral clamp	

13. Final Demo Checklist

Before Week 7:

- Arduino code runs without edits;
- Python recorder works;
- motor wiring is labeled or documented;
- final controller settings are recorded;
- plots are saved;
- demo reference speed is known;
- team members know their roles;
- backup CSV data are saved.

14. Check Your Understanding

Discuss with your team:

1. Which plot best proves that feedback improved your system?
2. Which controller setting had the biggest effect on your final result?
3. What is the main limitation of your current motor setup?
4. What will you do if the live demo does not behave exactly like your saved data?

Summary

- Final validation should be systematic and repeatable.
- Test more than one reference speed when practical.
- Use measured data to justify controller settings.
- Troubleshoot one variable at a time.
- The final presentation should tell the full open-loop-to-PI story.
- Week 7 is for presentations, demos, reflection, and final submission.